Field experiments

Lorenz was not alone in his theories. Other biologists working in the field included fellow Austrian Karl von Frisch and Dutch biologist Nikolaas Tinbergen, who studied animals in their natural environments. Until then, most animal behavior studies had taken place in laboratories or artificial settings, so the behavior witnessed was not entirely natural. Studying animals in the wild had its own challenges, particularly when devising rigorous field experiments that could be repeated, so that the findings could be recognized as facts, not anecdotes.

The term “ethology” was coined by American entomologist William Morton Wheeler in 1902 to describe the scientific study of animal behavior. Ethologists study animals in their natural habitats, combining laboratory studies and fieldwork in order to describe an animal’s behavior in relation to its ecology, evolution, and genetics.

Ethologists found that in certain situations, an animal will have a predictable behavioral response. They called this a “fixed action pattern” (FAP). A FAP has set characteristics. It is species-specific; it is repeated in the same way every time and is not affected by experience. The triggers for the behavior (“sign stimuli”) are highly specific and may involve a color, pattern, or sound. For example, male sticklebacks respond aggressively when another male enters their streambed patch. Ethologists suggest this is triggered by seeing the male’s red underbelly.

Nikolaas Tinbergen found that some artificial sign stimuli work better than the real thing. He investigated the begging behavior of herring gull chicks, which peck at a red spot on the parent gull’s beak to make it regurgitate food. He found that chicks will also peck at a model of the gull’s beak, yet when they were offered a narrow red pencil with three white lines at the end, the chicks pecked at this even more enthusiastically. Tinbergen